Dr. Alice Lacaze-Masmonteil

Postdoctoral Fellow at the University of Auckland

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Canadian and French Pronouns: She/her

Education

University of Ottawa **Ottawa, ON, Canada**
PhD in Mathematics and Statistics 2020-2024

Advisor: Mateja Šajna
Thesis: Certain resolvable directed cycle decompositions of directed graphs

Acadia University **Wolfville, NS, Canada**
Master of Science - Mathematics 2017-2019

Academic Positions

University of Auckland **Auckland, New Zealand**
Postdoctoral Fellow 2026-2029
Advisor: Florian Lehner

University of Regina **Regina, SK, Canada**
Pacific Institute for the Mathematical Sciences- Centre National de la Recherche Scientifique (PIMS-CNRS) Postdoctoral Fellow 2024-2026
Advisor: Karen Meagher

Research visits

University of Brescia **Brescia, Italy**
Visited Tommaso Traetta for 2 weeks. June 2026

Monash University **Melbourne, Australia**
Visited Daniel Horsley for 5 months. 2023

Research Interests

My main area of research is discrete mathematics. Namely, I am specialized in cycle decomposition problems.

Publications

Journal Articles Published

[1] H. Gupta, A. Herman, **A. Lacaze-Masmonteil**, R. Maleki, K. Meagher, On the second largest eigenvalue of certain graphs in the perfect matching association scheme, *Electronic Journal of Combinatorics*, **33(1)** (2026), 1–39.

[2] **A. Lacaze-Masmonteil**, Hamiltonian decompositions of the wreath product of hamiltonian decomposable digraphs, *Discrete Mathematics*, **349** (2026), 115012.

[3] D. Horsley and **A. Lacaze-Masmonteil**, Completing the solution of the directed Oberwolfach problem with two tables, *Electronic Journal of Combinatorics*, **33 (1)** (2026), 1–33.

[4] **A. Lacaze-Masmonteil**, Decompositions of the wreath product of certain directed graphs into directed hamiltonian cycles, *Ars Mathematicae Contemporanea*, **26 (2)** (2026)

[5] **A. Lacaze-Masmonteil**, Completing the solution of the directed Oberwolfach problem with cycles of equal length, *Journal of Combinatorial Designs*, **32** (2024), 5-30.

Journal Articles Under Review

[6] A. Herman, **A. Lacaze-Masmonteil**, K. Meagher, and H. Monterde, Near-factorizations in association schemes, submitted October 2026.

[7] A. C. Burgess, P. H. Danziger, **A. Lacaze-Masmonteil**, A complete solution to the directed Oberwolfach problem of order 2 (mod 4) with cycles of even lengths, submitted February 2026.

Fellowships, Prizes, and Scholarships

All monetary values are given in Canadian Dollar (CAD).

- Natural Sciences and Engineering Research Council of Canada (NSERC) Canada Postdoctoral Research Award (2027-2029)
  - Valued at $70,000 per year.
- University of Ottawa Department of Mathematics and Statistics PhD Thesis Award (2025)
  - Awarded to the best doctoral thesis in the department of mathematics and statistics.
- PIMS-CNRS Postdoctoral Fellowship (2024-2026)
  - Valued at $62,000 per year.
- University of Ottawa Mathematics and Statistics Outstanding Student Paper Prize (2023)
  - Awarded annually to the best paper published by a student.
- Peter Rodney Memorial Book Prize (2023)
  - Awarded to the best student talk given at the 27th Ontario Combinatorics Workshop.
- University of Ottawa André Dabrowski Scholarship Fund (2023)
  - Valued at $1,400. Awarded to a student that demonstrated qualities of leadership.
- NSERC Michael Smith Foreign Study Supplement (2023)
  - National research scholarship valued at $6,000 to fund visit to Monash University in Melbourne Australia.
- NSERC Canada Graduate Scholarship - Doctoral (2021-2024)
  - National research scholarship valued at $35,000 per year.
- Catherine Stanley Memorial Scholarship (2018)
  - Awarded to a student that demonstrated excellence as a teaching assistant.
- NSERC Canada Graduate Scholarship - Master's program (2017)
  - Institutional research scholarship valued at $17,500.

Mentorship

Semesters are given in terms of the standard North American academic calendar.

University of Regina **Regina, SK, Canada**
Undergraduate summer students mentor Summer 2025
Created a seminar series on scientific communication skills for undergraduate summer students in mathematics.

Teaching

University of Regina **Regina, SK, Canada**
Sessional Instructor 2024-Present
Courses (taught in english):
MATH 828: Combinatorics (a graduate class in design theory) Winter 2026
MATH 221: Introduction to Proofs and Problem Solving Fall 2025
MATH 223: Introduction to Abstract Algebra Winter 2025
MATH 122: Linear Algebra 1 Fall 2024

University of Ottawa **Ottawa, ON, Canada**
Part-time professor 2023
Courses (taught in French):
MAT 1741: Introduction to Linear Algebra Winter 2023

Presentations

Invited Conference Presentations

- 2025 Canadian Mathematical Society Winter Meeting: Algebraic Graph Theory Scientific Session, Toronto, ON, Canada, (Dec. 2025).
- 10th Cracow Conference on Graph Theory: Design Theory Session, Cracow, Poland, (Sep. 2025).
- Canadian Discrete and Algorithmic Mathematics Conference 2025: From Puzzles to Proofs: Connecting the Dots with Combinatorics, Ottawa, ON, Canada, (May 2025).
- 2024 Canadian Mathematical Society Winter Meeting: Combinatorial Designs Scientific Session, Vancouver, BC, Canada, (Dec. 2024).
- Banff International Research Station: Movement and Symmetry in Graphs Workshop, Banff, AB, Canada, (Nov. 2024).

- 10th Slovenian Conference on Graph Theory: Combinatorial Designs and their Applications Mini Symposium, Kranjska Gora, Slovenia, (June 2023).
- 2022 Canadian Mathematical Society Summer Meeting: Design Theory and Graph Decomposition Scientific Session, St. John's, NL, Canada, (June 2022).

Contributed Conference Presentations

- Combinatorics 2026, Naples, Italy, (May 2026).
- 2026 Prairie Discrete Mathematics Workshop, Regina, SK, Canada, (May 2026).
- 2025 Canadian Mathematical Society Winter Meeting: Combinatorial Designs Scientific Session, Toronto, ON, Canada, (Dec. 2025).
- Women in Combinatorics Virtual Conference, Online, (July 2024).
- 45th Australasian Combinatorics Conference, Perth, WA, Australia, (Dec. 2023).
- 2023 Canadian Mathematical Society Summer Meeting: Design Theory and Graph Decomposition Session, Ottawa, ON, Canada, (June 2023).
- 27th Ontario Combinatorics Workshop, Ottawa, ON, Canada, (May 2023).
- 26th Ontario Combinatorics Workshop, Waterloo, ON, Canada, (May 2022).
- 25th Ontario Combinatorics Workshop, Online, (June 2021).

Seminars

- University of Waterloo Algebraic Graph Theory Seminar, Online, (Apr. 2026).
- PIMS Emergent Research: The PIMS Postdoctoral Fellow Seminar, Online, (Feb. 2026).
- University of Manitoba Department of Mathematics Colloquium, Winnipeg, MB, Canada, (Nov. 2025).
- University of Victoria Discrete Mathematics Seminar, Victoria, BC, Canada, (Nov. 2025).
- PIMS Postdoctoral Summit, Calgary, AB, Canada, (Apr. 2025).
- AARMS Atlantic Graph Theory Seminar, Online, (Nov. 2024).
- Lethbridge Number Theory and Combinatorics Seminar Lethbridge, AB, Canada, (Oct. 2024).
- PIMS Emergent Research: The PIMS Postdoctoral Fellow Seminar, Online, (Oct. 2024).
- Monash University Discrete Mathematics Seminar, Melbourne, VIC, Australia, (Oct. 2023).

Selected Service

Referee for the following publications: Australasian Journal of Combinatorics, Journal of Combinatorial Designs, Graphs and Combinatorics, Bulletin of the Institute of Combinatorics and its Applications, Discussiones Mathematicae Graph Theory.

2026 Prairie Discrete Mathematics Workshop — 2026
Co-organizer

Math Kangaroo training camp (Grades 3 and 4) — 2026
Teacher

2025 Canadian Mathematical Society (CMS) Winter Meeting — 2025
Combinatorial Design Theory Session co-organizer

Canadian Discrete and Algorithmic Mathematics Conference 2025 — 2025
Women in combinatorics mentorship lunch co-organizer

Canadian Discrete and Algorithmic Mathematics Conference 2025 — 2025
Combinatorial Designs and Graph Decompositions Scientific Session co-organizer

Pi Day at the Saskatchewan Science Centre — 2025
Facilitator

University of Regina Day - Mathematics Outreach — 2024
Facilitator

Women in Combinatorics Virtual Conference — 2024
Combinatorial Designs Session co-organizer

2023 CMS Summer Meeting — 2023
Design Theory and Graph Decomposition Session co-organizer

27th Ontario Combinatorics Conference — 2023
Co-organizer

Canadian Mathematical Society Student Committee — 2022-2024
Chair

UO Mathematics and Statistics Graduate Student Association Executive — 2022-2023

President
Ottawa Mathematics and Statistics Conference 2021-2023
Co-organizer

References

- Karen Meagher, University of Regina
- Daniel Horsley, Monash University
- Mateja Šajna, University of Ottawa
- Shaun Fallat, University of Regina, (Teaching/service reference)